

Math 241, F14
Quiz 5

Name: *Answer*
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Let $\mathbf{r}(t) = \langle 5t, 2\sin t, 2\cos t \rangle$. Find the curvature of the curve using the formula $\kappa = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$.

$$\mathbf{r}'(t) = \langle 5, 2\cos t, -2\sin t \rangle, \mathbf{r}''(t) = \langle 0, -2\sin t, -2\cos t \rangle$$

$$\begin{aligned} \mathbf{r}'(t) \times \mathbf{r}''(t) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 2\cos t & -2\sin t \\ 0 & -2\sin t & -2\cos t \end{vmatrix} = \mathbf{i}(-4\cos^2 t - 4\sin^2 t) - \mathbf{j}(-10\cos t) \\ &\quad + \mathbf{k}(-10\sin t) \\ &= -4\mathbf{i} + (10\cos t)\mathbf{j} - (10\sin t)\mathbf{k} \end{aligned}$$

$$|\mathbf{r}'(t)| = \sqrt{5^2 + (2\cos t)^2 + (-2\sin t)^2} = \sqrt{25 + 4\cos^2 t + 4\sin^2 t} = \sqrt{25 + 4} = \sqrt{29}$$

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = \sqrt{(-4)^2 + (10\cos t)^2 + (-10\sin t)^2} = \sqrt{16 + 100\cos^2 t + 100\sin^2 t} = \sqrt{116} = 2\sqrt{29}$$

$$\Rightarrow \kappa = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3} = \frac{2\sqrt{29}}{(\sqrt{29})^3} = \frac{2}{29}$$

Problem 2. (5 points) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has $\mathbf{a}(t) = \langle t, t^2, 1 \rangle$, $\mathbf{v}(0) = \langle 0, 0, 1 \rangle$ and $\mathbf{r}(0) = \langle 1, 0, 0 \rangle$.

$$\mathbf{v}(t) = \mathbf{v}(0) + \int_0^t \mathbf{a}(u) du = \langle 0, 0, 1 \rangle + \int_0^t \langle u, u^2, 1 \rangle du$$

$$= \langle 0, 0, 1 \rangle + \langle \frac{t^2}{2}, \frac{t^3}{3}, t \rangle$$

$$= \langle \frac{t^2}{2}, \frac{t^3}{3}, t+1 \rangle$$

$$\mathbf{r}(t) = \mathbf{r}(0) + \int_0^t \mathbf{v}(u) du = \langle 1, 0, 0 \rangle + \int_0^t \langle \frac{u^2}{2}, \frac{u^3}{3}, u+1 \rangle du$$

$$= \langle 1, 0, 0 \rangle + \langle \frac{t^3}{6}, \frac{t^4}{12}, \frac{t^2}{2} + t \rangle$$

$$= \langle \frac{t^3}{6} + 1, \frac{t^4}{12}, \frac{t^2}{2} + t \rangle$$